

① 5.2 -16

$$F_n = \begin{vmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & -1 & 1 \end{vmatrix}$$

$$F_n = \begin{vmatrix} \textcircled{1} & \textcircled{-1} & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & -1 & 1 \end{vmatrix} \rightarrow F_{n-1}$$

$$F_n = F_{n-1} + (-1)^{1+2} (-1) (F_{n-2})$$

$$\begin{vmatrix} 1 & \textcircled{-1} \\ \textcircled{1} & 1 & \textcircled{-1} & \dots \\ 0 & 0 & 1 & \dots \end{vmatrix} \rightarrow F_{n-2}$$

$$F_n = F_{n-1} + F_{n-2}$$

② $S \cdot 2 - 32$

$$S_n = \left| \begin{bmatrix} 3 & 1 & & \\ 1 & 3 & & \\ & 1 & 3 & \\ & & 1 & \ddots \end{bmatrix} \right|$$

$$S_n = 3 S_{n-1} - S_{n-2}$$

$$\hookrightarrow F_{2n+2} = F_{2n+1} + F_{2n}$$

$$= 2F_{2n} + F_{2n-1}$$

$$F_{2n} = F_{2n-1} + F_{2n-2}$$

$$F_{2n-1} = F_{2n} - F_{2n-2}$$

$$F_{2n+2} = 3F_{2n} - F_{2n-2}$$

③

S.2-33

$$\begin{vmatrix} \ddots & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \ddots \end{vmatrix}_{20} = \begin{vmatrix} \ddots & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \ddots \end{vmatrix}_{19} + \begin{vmatrix} \ddots & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \ddots \end{vmatrix}_{19} \begin{matrix} \nearrow P_{33}=1 \\ \\ \\ \textcircled{1} \end{matrix}$$

$$\det \underbrace{(P_4)}_1 = \boxed{\underbrace{\det(X)}_0} + \underbrace{\det(P_3)}_1$$

$$\textcircled{4} \quad \underline{\underline{5 \cdot 3 - 8}}$$

$$C_{11} = 6$$

$$C_{12} = -3$$

$$\boxed{C_{13} = 0} \rightarrow \text{this is why } \det A \text{ is unchanged} \checkmark$$

$$A = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \quad C^T = \begin{bmatrix} 6 & \cdot & \cdot \\ -3 & \cdot & \cdot \\ 0 & \cdot & \cdot \end{bmatrix}$$

$$AC^T = \begin{bmatrix} 3 & & \\ & 3 & \\ & & 3 \end{bmatrix} = \underline{\underline{3I}}$$

⑤ 5.3 - 28

$\Rightarrow \rho, \phi, \theta \rightarrow$ spherical co-ordinates

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\left(\frac{\partial}{\partial \rho} \right) = \begin{bmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{bmatrix}$$

$$\det() = J = \rho^2 \cos \phi \begin{bmatrix} \sin \phi & \cos \phi \end{bmatrix}$$

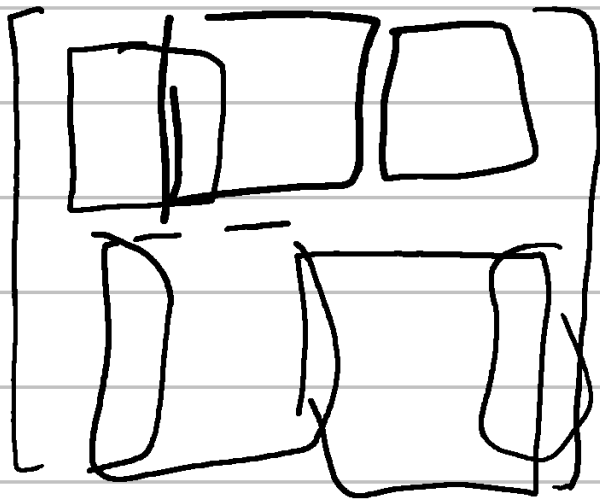
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$$\rho^2 \sin \phi \begin{bmatrix} \sin^2 \phi \end{bmatrix}$$

$$\Rightarrow \rho^2 \sin \phi$$

⑥ 5-3-46

A is 5×5 matrix



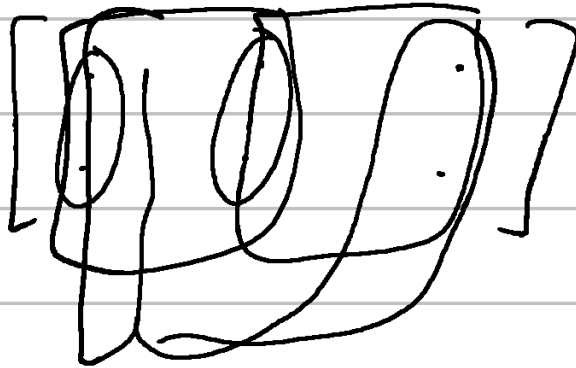
$$\begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}$$

$$3 \cdot 4 = 6$$

⑦ 5.3 - 41

$$AB = (2 \times 3) (3 \times 2)$$

a)



3C_2 ways

29 & 16

b)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & ? \end{bmatrix}$$

$B =$

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & ? \end{bmatrix}$$

$$\det(A)'s = \begin{array}{l} \text{cols} \\ (1-2) = 2 \\ (1-3) = 4 \\ (2-1) = 2 \end{array}$$

$$\det(B)'s = \begin{array}{l} (1-2) = 2 \\ (1-3) = 4 \\ (2-1) = 2 \end{array}$$

$$AA^T = \begin{bmatrix} 14 & 30 \\ 30 & 66 \end{bmatrix}$$

$$4 + 16 + 4$$

(20)

$$66 \cdot 14 - 400 = \underline{22}$$

⑧ 6.1-19

$$B \rightarrow 3 \times 1$$

$$\lambda \rightarrow 0, 1, 2$$

$$B \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$a) \text{rank}(B) = 2$$

$$b) \det(B^T B) = \underline{\underline{0}}$$

$$c) \lambda \rightarrow B^T B \rightarrow$$

$$d) (B^2 + I)^{-1}$$

$$\lambda \rightarrow 0, 1, 4$$

$$B^2 + I \rightarrow 1, 2, 5$$

$$1, \frac{1}{2}, \frac{1}{5}$$

⑨ 6.1-29

$$A \Rightarrow 1, 4, 6$$

$$B \rightarrow \begin{bmatrix} -\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 3 & 0 & -\lambda \end{bmatrix}$$

$$\bullet \lambda (\bullet \lambda \cdot (2-\lambda)) + 1 (-3 \cdot (2-\lambda)) = 0$$

$$\lambda^2 (2-\lambda) - 3 (2-\lambda)$$

$$(2-\lambda) (\lambda + \sqrt{3}) (\lambda - \sqrt{3}) = 0$$

$$\lambda = 2, \sqrt{3}, -\sqrt{3}$$

$$C \rightarrow \begin{bmatrix} 2-\lambda & 2 & 4 \\ 2 & 2-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{bmatrix}$$

$$(2-\lambda) ((2-\lambda)^2 - 4) - 2(2 \cdot (2-\lambda) - 4) + 2(4 - 2 \cdot (2-\lambda))$$

$$(2-\lambda) (\cancel{4} + \lambda^2 - 4\lambda - \cancel{4}) - 2(\cancel{4} - 2\lambda - \cancel{4}) + 2(\cancel{4} - \cancel{4} + 2\lambda)$$

$$(2-\lambda) (\lambda(\lambda-4)) + 8\lambda \Rightarrow \lambda(2\lambda + 4\lambda - \cancel{\lambda^2} - \cancel{8\lambda}) = 0$$

$$\boxed{\lambda^2 (\lambda - 6) = 0}$$

⑩ 6.2 - 6

S that diagonalize A

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}$$

$$A = S^{-1} \Lambda S$$
$$A^{-1} = (S^{-1} \Lambda S)^{-1}$$

$= S^{-1} \Lambda S$
Same!

$$(4 - \lambda)(2 - \lambda) = 0$$

$\lambda = 2, 4$

$$\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}$$

$$\lambda_1 = 2 \quad x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2 - 2 = 0$$

$$\lambda_2 = 4 \quad x_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} | & | \\ c_1 x_1 & c_2 x_2 \\ | & | \end{bmatrix}$$

① 6.2-16

$$A_1 = \begin{bmatrix} .6 & .9 \\ .4 & .1 \end{bmatrix}$$

$$\lambda = 1, -0.3$$

$$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -0.3 \end{bmatrix}$$

$$\begin{bmatrix} -0.4 & 0.9 \\ 0.4 & -0.9 \end{bmatrix}$$

$$\lambda_1 = 1 \quad x_1 = \begin{bmatrix} a \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} -0.4 & 0.9 \\ 0 & 0 \end{bmatrix}$$

$$\lambda_2 = -0.3 \quad \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0.9 & 0.9 \\ 0.4 & 0.4 \end{bmatrix}$$

$$S = \begin{bmatrix} a & -1 \\ 4 & 1 \end{bmatrix}$$

$$S \Lambda^k S^{-1}$$

$$\Lambda^k \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ as } k \rightarrow \infty$$

$$S = \begin{bmatrix} 9 & -1 \\ 4 & 1 \end{bmatrix}$$

$$\Lambda^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S^{-1} = \frac{1}{13} \begin{bmatrix} 1 & 1 \\ -4 & 9 \end{bmatrix}$$

$$S \Lambda^2 S^{-1} = \frac{1}{13} \begin{bmatrix} 9 & 9 \\ 4 & 4 \end{bmatrix}$$

② 6.2-33

$$A = S \Lambda S^{-1}$$

$$A^T = (S^{-1})^T \Lambda S^T$$

$x_1, x_2, \dots, x_n$ in $A^T y = \lambda y$ left
sum of rank 1 matrices

$$A = S \Lambda S^{-1} = \lambda_1 x_1 y_1^T + \dots + \lambda_n x_n y_n^T.$$

$$(S^{-1})^T = \begin{bmatrix} | & | & & | \\ y_1 & y_2 & \dots & y_n \\ | & | & & | \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} -y_1- \\ \vdots \\ -y_n- \end{bmatrix}$$

$$\begin{bmatrix} \begin{pmatrix} | \\ x_1 \\ | \end{pmatrix} \dots \begin{pmatrix} | \\ x_n \\ | \end{pmatrix} \end{bmatrix} \begin{bmatrix} -\lambda_1 y_1- \\ -\lambda_2 y_2- \\ \vdots \end{bmatrix}$$

$$\lambda_1 x_1 y_1^T \dots$$